



ECG Arrhythmia Classification with Real-time Hospital Monitoring

Powered by RhythmNet AI Technology

Advanced Machine Learning Workshop: From Research to Clinical Implementation



Real-time Monitoring



AI-Powered



98.54% Accuracy

Introduction: The Clinical Problem

Global Burden of Cardiac Arrhythmias



15-20%

of global deaths

Cardiac arrhythmias contribute to deaths worldwide, primarily through sudden cardiac arrest



1-5 min

per patient

Manual ECG interpretation time in busy clinical settings, creating bottlenecks

Critical Impact Areas



Sudden cardiac arrest - leading cause



Emergency department overload



Limited cardiologist availability



Time-critical diagnosis requirements

Introduction: The Clinical Problem

Interpretation Challenges & Urgency



ECGs Daily

Emergency departments handle massive volumes with limited specialist coverage

! **10-15%**

Misinterpretation

Error rates in non-specialist settings, especially dangerous for critical arrhythmias



V-Tach Detection

Ventricular tachycardia requires immediate detection - delays directly impact survival

Business Motivation & Market Impact

Market Opportunity & Healthcare Costs

 **\$10B+**
Annual Market

Global cardiac monitoring device market with rapid growth trajectory

Market Drivers

- Aging population demographics
- Increasing cardiovascular disease prevalence
- Technology adoption in healthcare



Hundreds of Billions

Global CV care costs

Significant portion from diagnostic delays and inefficiencies



Rural Hospitals

Coverage gaps

Lack 24/7 cardiologist coverage, creating diagnostic gaps



Key Insight

AI-powered diagnostic assistance represents major market opportunity for healthcare cost reduction

Business Motivation & Market Impact

AI's Role in Addressing Critical Gaps

Current Challenges



False Negatives

Life-threatening delays and malpractice liability



Alert Fatigue

High false positive rates in current systems



Coverage Gaps

Rural hospitals lack 24/7 specialist availability

AI-Powered Solution



Intelligent Assistance

24/7 diagnostic support for all healthcare settings



Cost Reduction

Major market opportunity for healthcare efficiency



Patient Safety

Reduced diagnostic errors and improved outcomes

The AI Opportunity

Transform cardiac care with intelligent, real-time diagnostic assistance

Bridge the Gap

Technology meets clinical need

Our Proposed Solution: RhythmNet

Core Features & Performance



98.54%

Accuracy

High Precision

Real-time ECG arrhythmia classification with <1 second processing time



47

Patients

Simultaneous Monitoring

Continuous tracking with intelligent alert prioritization system



Smart

Alerts

Clinical Decision Support

Interpretable AI outputs with confidence scores and smart alerting

Performance Breakthrough

- Sub-second processing time for real-time analysis
- Intelligent alert algorithm reduces false alarms
- Seamless integration with existing hospital infrastructure

< 1s

Processing Time

Real-time clinical decision support

Our Proposed Solution: RhythmNet

Advanced Capabilities & Integration



Live Visualization

Continuous monitoring without workflow disruption, seamless clinical integration.



Interpretability

Interpretable AI outputs providing clinical decision support with transparency.



Patient Prioritization

Abnormal patients automatically moved to top of clinical dashboard with visual highlighting.

Development Philosophy & Methodology



Clinical-First Approach

Prioritizing sensitivity and specificity over raw accuracy metrics for patient safety.



Advanced Class Balancing

SMOTE techniques specifically optimized for rare but critical arrhythmia detection.



Comprehensive Comparison

Traditional ML and state-of-the-art deep learning approaches for robust validation.

CRISP-DM Methodology Framework I

Phases 1-3: Foundation & Preparation

Systematic Data Science Approach

Industry-standard methodology ensuring rigorous research and development

CRISP-DM

6-Phase Framework

CRISP-DM Methodology Framework II

Phases 4-6: Modeling to Deployment



Phase 4

Modeling

Systematic Evaluation

- 5 different architectures
- Traditional ML to transformers
- RhythmNet development
- Cross-validation testing



Phase 5

Evaluation

Rigorous Validation

- Clinical metrics focus
- Sensitivity & specificity
- Per-class performance
- Real-world testing



Phase 6

Deployment

Real-time Implementation

- Hospital system integration
- Clinical interface design
- Performance validation

RhythmNet Architecture

Hybrid CNN-Attention Architecture



CNN Local Patterns

Convolutional layers for local feature extraction and pattern recognition



Attention Global Context

Self-attention mechanism for global temporal modeling



P-QRS-T Detection

Automated detection of cardiac waveform components



Interpretable Output

Attention maps provide clinical explanations



Multi-Scale

Multiple kernel sizes for comprehensive feature extraction



Long-Range

Captures dependencies across entire heartbeat sequence



Adaptive

Channel attention emphasizes discriminative features

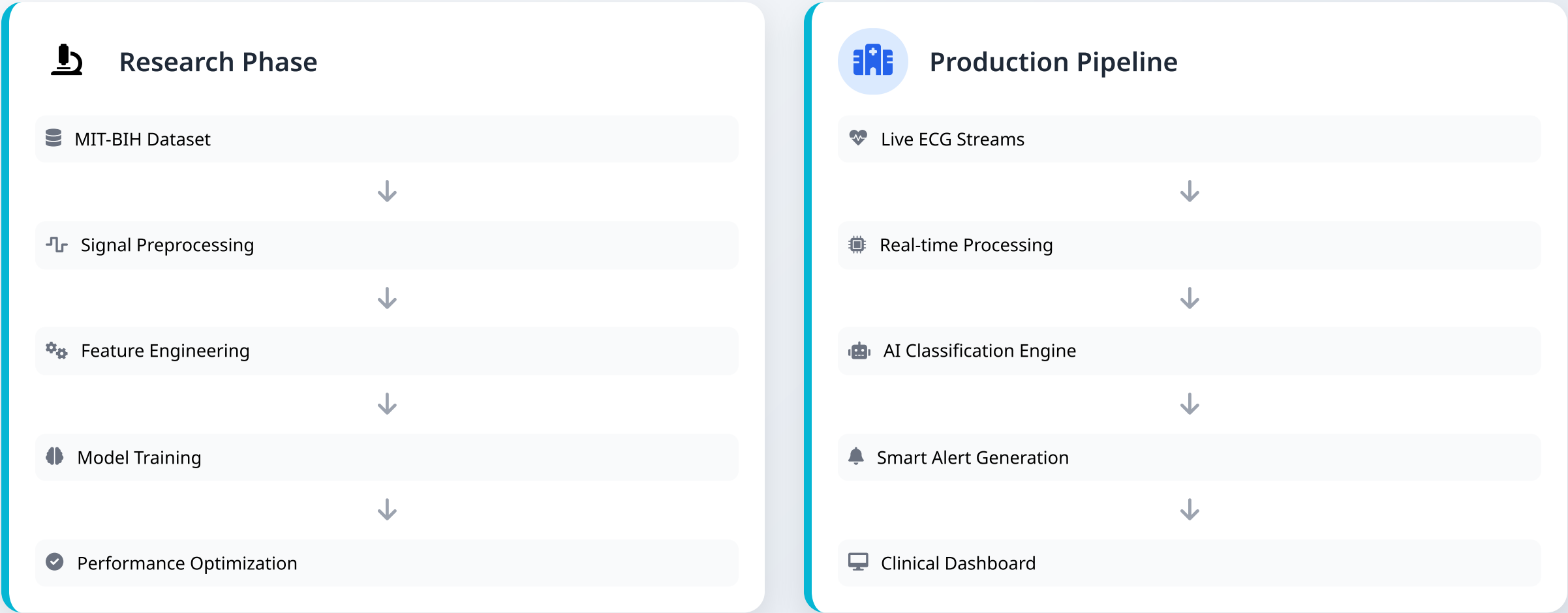


Efficient

Optimized for real-time clinical deployment

Technical Pipeline Architecture

Research & Production Workflows



Technical Pipeline Architecture

Integration & Quality Assurance



Hospital ECG Machines

Direct integration with existing ECG monitoring infrastructure



Real-time AI Engine

Local processing with RhythmNet model for instant ECG classification



Patient Management

Integrated clinical decision support interface

MIT-BIH Arrhythmia Database

Gold Standard Dataset Overview



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Patients

Patient Population

Diverse demographics with various cardiac conditions and noise levels



87,554

Samples

Training Data

Expert cardiologist annotations following AHA guidelines



187

Values

Signal Structure

ECG values per heartbeat (1.496s windows at 125Hz)

Gold Standard Quality

- Zero missing values in dataset
- Pre-normalized signal amplitudes
- Consistent temporal alignment around R-peaks
- Expert cardiologist validation



Industry Standard

Trusted by researchers worldwide

MIT-BIH Arrhythmia Database

Classification Categories & Validation

N

Normal

Regular QRS complexes, consistent PR intervals, stable rhythm

S

Supraventricular

Premature atrial contractions requiring monitoring

V

Ventricular

Potentially life-threatening premature ventricular contractions

F

Fusion

Rare overlapping electrical activity, clinically significant

Q

Unclassifiable

Noisy or ambiguous signals challenging for experts

Clinical Validation

- Expert cardiologist annotations
- American Heart Association guidelines
- Representative patient demographics
- Various cardiac conditions included

5

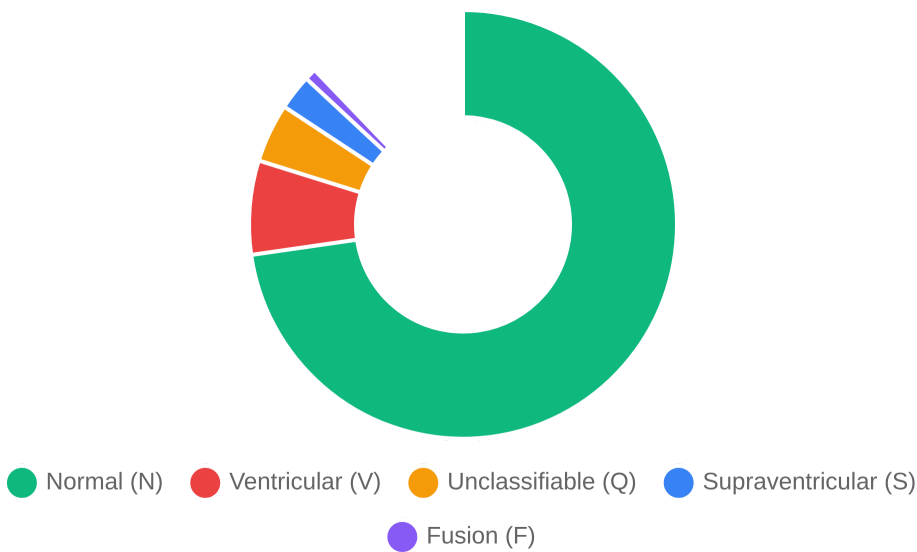
Classification Categories

Comprehensive arrhythmia coverage

Class Distribution Challenge

Real-World Clinical Imbalance

Sample Distribution



Critical Clinical Implication

Missing minority classes can result in life-threatening diagnostic errors

N

Normal

82%

~72,000 samples - majority class representing healthy rhythm

V

Ventricular

8%

~7,000 samples - potentially life-threatening PVCs

Q

Unclassifiable

5%

~4,700 samples - noisy or ambiguous signals

S

Supraventricular

3%

~3,000 samples - premature atrial contractions

F

Fusion

1%

Advanced Feature Engineering

Multi-Domain Signal Analysis

Statistical

- Mean amplitude
- Standard deviation
- Skewness
- Kurtosis
- Signal distribution

Morphological

- R-peak detection
- QRS width measurement
- P-wave identification
- T-wave analysis
- Waveform shape

Temporal

- RR interval variability
- Heart rate calculation
- Rhythm regularity
- Beat-to-beat timing
- Sequence patterns

Frequency

- FFT components
- Power spectral density
- Dominant frequency
- Spectral analysis
- Frequency bands

Entropy

- Sample entropy
- Approximate entropy
- Permutation entropy
- Signal complexity
- Irregularity measures

Wavelet

- Multi-resolution decomposition
- Time-frequency analysis
- Wavelet coefficients
- Scale-based features
- Localized patterns

Class Balancing Strategy & SMOTE Implementation

Addressing Critical Diagnostic Bias



Core Problem

82% Normal beats vs 3% Supraventricular vs 8% Ventricular creates severe learning bias

Missing rare arrhythmias = life-threatening diagnostic errors



SMOTE Solution

Synthetic Minority Oversampling Technique generating realistic synthetic ECG patterns

K-nearest neighbors approach preserving class characteristics

Implementation & Validation



K-Nearest Neighbors

Interpolated samples preserving morphological characteristics



Stratified Validation

Original distribution preserved in evaluation sets



Quality Control

Synthetic samples validated against clinical morphology criteria

Performance Improvement

Minority class sensitivity significantly enhanced while maintaining overall accuracy

86.76% → 90.54%
Minority Class Sensitivity

Comprehensive Model Comparison

Architecture Evaluation Framework



RhythmNet (Hybrid)

CNN feature extraction + attention-based temporal modeling



ViT-1D Transformer

Vision transformer adaptation applying self-attention to 1D ECG sequences



XGBoost + SMOTE

Traditional machine learning with gradient boosting and manually engineered features



Engineered Features

Comparing raw signal input vs manually crafted statistical and morphological features

Evaluation Framework



Clinical Metrics Priority

Sensitivity, specificity, and per-class performance over raw accuracy



Cross-Validation

5-fold stratified validation ensuring robust performance estimation



Patient Populations

Diverse demographics and cardiac conditions representation

Key Insight

Deep learning architectures superior for complex temporal pattern recognition in ECG signals

Detailed Model Performance Comparison

Comprehensive Metrics Analysis

Model Architecture	Overall Accuracy	Macro F1-Score	Sensitivity	Specificity	Training Time
 RhythmNet (Hybrid)	98.54%	92.33%	91.03%	98.89%	550.5 seconds
 ViT-1D Transformer	98.28%	91.47%	89.99%	98.70%	577.7 seconds
 XGBoost + SMOTE	98.29%	90.66%	90.54%	98.70%	210.8 seconds
 Engineered Features	98.29%	90.66%	86.76%	98.34%	470.3 seconds

Clinical Relevance

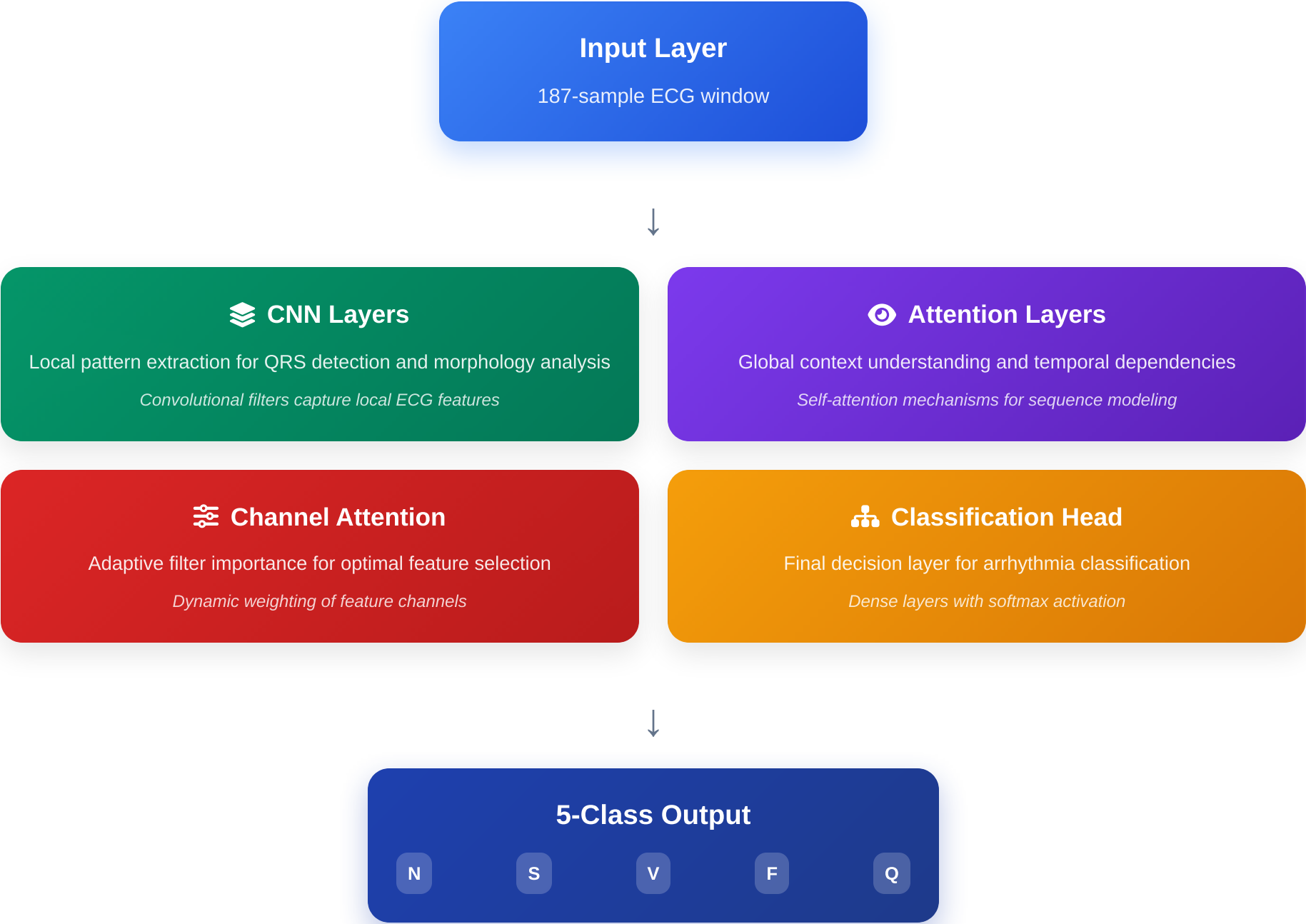
- RhythmNet achieves optimal balance of accuracy and efficiency
- Deep learning superior for complex sequential pattern recognition
- Clinical metrics prioritized over raw accuracy



Key Insight

RhythmNet: Best overall performance

RhythmNet Architecture Visualization

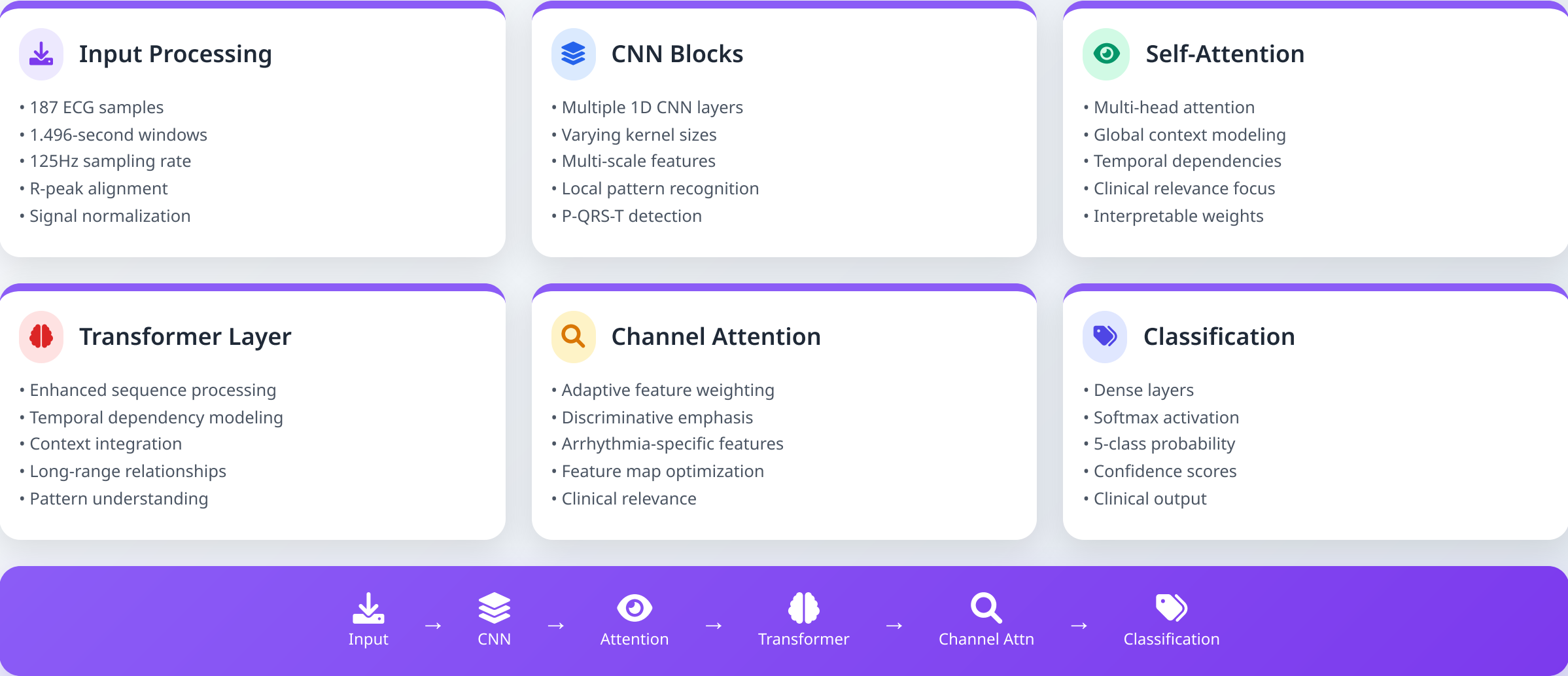


Hybrid Innovation

RhythmNet combines the best of CNN and Transformer architectures: CNNs excel at local pattern recognition (QRS complexes, wave morphology) while attention mechanisms capture global temporal dependencies and long-range relationships in ECG signals.

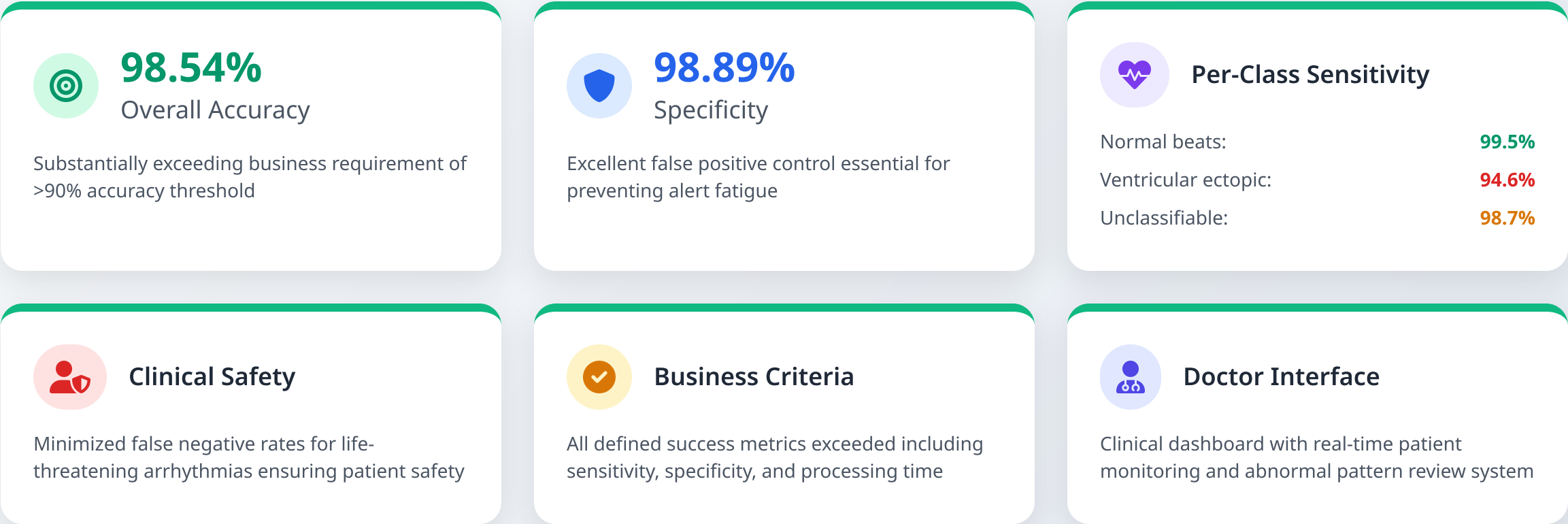
RhythmNet Architecture Deep Dive

Core Components & Processing Flow



Clinical Performance Validation

Real-World Performance Metrics



Future Directions

Research Extensions

Prospective Clinical Validation

Large-scale prospective studies with diverse patient populations to validate real-world clinical effectiveness.

Multi-site randomized controlled trials and outcome studies

Multi-center Hospital Trials

Collaborative research across multiple hospital systems to assess generalizability and clinical impact.

International collaboration and diverse healthcare settings

Vital Signs Integration

Combining ECG analysis with blood pressure, oxygen saturation, and other vital signs for comprehensive monitoring.

Multimodal AI for holistic patient assessment

Predictive Analytics

Developing predictive models for cardiac events and risk stratification based on ECG patterns.

System Scalability

Cloud Deployment

Scalable cloud infrastructure for hospital networks and healthcare systems with auto-scaling capabilities.

AWS, Azure, and GCP deployment with global availability

Edge Computing

Edge deployment for real-time processing with minimal latency and reduced bandwidth requirements.

Local processing for critical care environments

International Standards

Compliance with international medical device standards (ISO 13485, IEC 62304) for global deployment.

CE marking, Health Canada, and other regulatory approvals

Federated Learning

Privacy-preserving federated learning for model improvement across institutions without data sharing.

Thank You!

Questions & Discussion



Real-time ECG Arrhythmia Classification Project

From CRISP-DM methodology to clinical deployment, we've successfully developed and implemented a comprehensive AI solution that bridges the gap between research innovation and real-world healthcare applications.

98.54%

Model Accuracy

< 1 sec

Processing Speed

5

Arrhythmia Classes

24/7

Real-time Operation



Questions & Discussion

We welcome your questions about our methodology, technical implementation, clinical validation, or future research directions.

Project Team

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